

Reverse causality and omitted variables in the shape-AoA association (Study 4)

Lexical statistics project

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This document addresses the reverse-causality and omitted-variable concern for the Study 4 result (the shape rating predicts earlier AoA): adult raters may judge a noun shape-organized partly *because* it is a familiar, early, highly imageable word, rather than the lexical statistic driving acquisition. The check is to refit the Table 3 model with covariates that index the “familiar early word” pathway: adult imageability and familiarity norms (Glasgow Norms; Scott et al., 2019), adult English AoA norms (Kuperman, Stadthagen-Gonzalez, & Brysbaert, 2012), and the English-anchored item norms used in the Wordbank prediction literature (babi-ness, valence, arousal; Braginsky et al., 2019). If the shape coefficient survives these covariates, a pure reverse-causality account becomes harder to maintain; if it does not, the association cannot be separated from adult lexical familiarity. Because Braginsky et al.’s language-specific covariates (MLU, phoneme counts) exist for only four of our fourteen languages, we use their English-anchored item-level norms, which enter exactly the way concreteness already does in the manuscript.

Data

```
# Primary Study 4 analysis set (one row per unilemma-by-language observation).
ratings_aoa <- read_csv(here("data/predictors", "ratings_aoa.csv"),
                        show_col_types = FALSE)

# Unilemma cleaning for matching against single-word English norms:
# lowercase, strip disambiguating parentheticals ("chicken (animal)"),
# and take the first alternative of slash-separated forms ("soda/pop").
clean_lemma <- function(x) {
  x %>% str_to_lower() %>%
    str_remove("\\s*\\(.*\\)\\s*$") %>%
    str_split_fixed("/", 2) %>% .[, 1] %>%
    str_trim()
}
ratings_aoa <- ratings_aoa %>% mutate(match_word = clean_lemma(uni_lemma))

# Kuperman et al. (2012) adult AoA ratings (downloaded from osf.io/d7x6q).
kuperman <- read_xlsx(here("data/predictors/external",
                           "kuperman_aoa_with_pos.xlsx")) %>%

  clean_names() %>%
  transmute(match_word = str_to_lower(word),
            adult_aoa = as.numeric(rating_mean)) %>%
  filter(!is.na(adult_aoa)) %>%
  distinct(match_word, .keep_all = TRUE)
```

```

# Glasgow Norms (Scott et al., 2019): imageability and familiarity
# (downloaded from the JASP data library mirror of the BRM supplement).
glasgow <- read_csv(here("data/predictors/external", "glasgow_norms.csv"),
  show_col_types = FALSE) %>%
  clean_names() %>%
  transmute(match_word = clean_lemma(word),
    imageability = as.numeric(m_imag),
    familiarity = as.numeric(m_fam)) %>%
  group_by(match_word) %>%
  summarise(across(c(imageability, familiarity), mean), .groups = "drop")

# Braginsky et al. (2019) English-anchored item norms, extracted from the
# aoa-prediction pipeline (uni_model_data.RData), English (American) rows.
braginsky <- read_csv(here("data/predictors/external",
  "braginsky_uni_predictors.csv"),
  show_col_types = FALSE) %>%
  filter(language == "English (American)") %>%
  transmute(match_word = clean_lemma(uni_lemma),
    babiness, valence, arousal) %>%
  distinct(match_word, .keep_all = TRUE)

model_df <- ratings_aoa %>%
  left_join(kuperman, by = "match_word") %>%
  left_join(glasgow, by = "match_word") %>%
  left_join(braginsky, by = "match_word") %>%
  distinct(language, uni_lemma, .keep_all = TRUE)

coverage <- tibble(
  covariate = c("adult AoA (Kuperman)", "imageability (Glasgow)",
    "familiarity (Glasgow)", "babiness (Braginsky)"),
  n_unilemmas = c(
    n_distinct(model_df$uni_lemma[!is.na(model_df$adult_aoa)]),
    n_distinct(model_df$uni_lemma[!is.na(model_df$imageability)]),
    n_distinct(model_df$uni_lemma[!is.na(model_df$familiarity)]),
    n_distinct(model_df$uni_lemma[!is.na(model_df$babiness)])),
  n_total = n_distinct(model_df$uni_lemma))
knitr::kable(coverage, caption = "Norm coverage of the Study 4 unilemma set.")

```

Table 1: Norm coverage of the Study 4 unilemma set.

covariate	n_unilemmas	n_total
adult AoA (Kuperman)	588	655
imageability (Glasgow)	413	655
familiarity (Glasgow)	413	655
babiness (Braginsky)	203	655

Models

Each augmented model is compared against the Table 3 baseline refit on the same rows (complete cases for that covariate set), so coefficient changes reflect the added covariates rather than the changed sample.

```

z <- function(x) as.numeric(scale(x))

fit_pair <- function(df, extra_terms, label) {
  covs <- c("logfreq", "concreteness", "solid", "mean_shape", "count_noun",
            extra_terms)
  d <- df %>%
    select(uni_lemma, aoa, category, language, all_of(covs)) %>%
    drop_na() %>%
    mutate(across(all_of(covs), z)) %>%
    distinct()
  base_form <- aoa ~ logfreq + concreteness + solid + mean_shape +
    count_noun + language + (1 | category) + (1 | uni_lemma)
  aug_form <- update(base_form,
                    paste(". ~ . +", paste(extra_terms, collapse = " + ")))
  fit_b <- lmerTest::lmer(base_form, data = d)
  fit_a <- lmerTest::lmer(aug_form, data = d)
  cb <- coef(summary(fit_b))["mean_shape", ]
  ca <- coef(summary(fit_a))["mean_shape", ]
  tibble(
    model = label,
    n_rows = nrow(d),
    n_unilemma = n_distinct(d$uni_lemma),
    shape_b_baseline = cb[["Estimate"]], shape_p_baseline = cb[["Pr(>|t|)"]],
    shape_b_augmented = ca[["Estimate"]], shape_p_augmented = ca[["Pr(>|t|)"]],
    extra = list(coef(summary(fit_a))[extra_terms, , drop = FALSE]))
}

res <- bind_rows(
  fit_pair(model_df, c("imageability", "familiarity"),
            "+ imageability + familiarity"),
  fit_pair(model_df, "adult_aoa", "+ adult English AoA"),
  fit_pair(model_df, c("babiness", "valence", "arousal"),
            "+ babiness + valence + arousal"),
  fit_pair(model_df, c("imageability", "familiarity", "adult_aoa",
                      "babiness", "valence", "arousal"),
            "kitchen sink (all six)"))

res %>%
  select(-extra) %>%
  mutate(across(where(is.numeric), ~ round(.x, 4))) %>%
  knitr::kable(caption = paste(
    "Shape coefficient (months of AoA per SD of shape rating) before and",
    "after adding the covariates, on identical rows.))

```

Table 2: Shape coefficient (months of AoA per SD of shape rating)
before and after adding the covariates, on identical rows.

model	n_rows	n_unilemma	shape_b_baseline	shape_p_baseline	shape_b_augmented	shape_p_augmented
+ imageability + familiarity	2533	413	-0.3088	0.0317	-0.3445	0.0110
+ adult English AoA	3216	588	-0.2864	0.0152	-0.2861	0.0085

model	n_rows	n_lemma	shape_b_base	shape_p_base	shape_b_augmented	shape_p_augmented
+ babiness + valence	1888	203	-0.2852	0.0988	-0.2753	0.1042
+ arousal						
kitchen sink (all six)	1531	154	-0.3304	0.0933	-0.3456	0.0411

Coefficients of the added covariates in each augmented model.

```
for (i in seq_len(nrow(res))) {
  cat("\n##", res$model[i], "\n")
  print(round(res$extra[[i]], 4))
}
```

```
##
## ## + imageability + familiarity
##           Estimate Std. Error      df t value Pr(>|t|)
## imageability  0.1407      0.1411 401.1188  0.9971  0.3193
## familiarity   -0.8399      0.1128 340.8181 -7.4448  0.0000
##
## ## + adult English AoA
##           Estimate Std. Error      df t value Pr(>|t|)
## adult_aoa     0.6966      0.071 713.314  9.8073      0
##
## ## + babiness + valence + arousal
##           Estimate Std. Error      df t value Pr(>|t|)
## babiness    -0.5705      0.1402 185.2526 -4.0683  0.0001
## valence      0.0615      0.1514 184.6320  0.4060  0.6852
## arousal      0.0649      0.1467 182.1451  0.4422  0.6589
##
## ## kitchen sink (all six)
##           Estimate Std. Error      df t value Pr(>|t|)
## imageability  0.2911      0.1802 145.3121  1.6157  0.1083
## familiarity   -0.4027      0.1688 139.0716 -2.3859  0.0184
## adult_aoa     0.9185      0.1565 141.3881  5.8693  0.0000
## babiness     -0.2266      0.1581 136.6190 -1.4333  0.1541
## valence       0.2514      0.1615 134.9793  1.5567  0.1219
## arousal       0.0506      0.1478 134.8022  0.3424  0.7326
```

Range restriction of the adult norms in this item set

```
rr <- bind_rows(
  tibble(norm = "adult AoA (Kuperman)",
    sd_our_items = sd(model_df %>% distinct(uni_lemma, adult_aoa) %>%
      pull(adult_aoa), na.rm = TRUE),
    sd_full_norms = sd(kuperman$adult_aoa)),
  tibble(norm = "imageability (Glasgow)",
    sd_our_items = sd(model_df %>% distinct(uni_lemma, imageability) %>%
      pull(imageability), na.rm = TRUE),
    sd_full_norms = sd(glasgow$imageability, na.rm = TRUE)),
  tibble(norm = "familiarity (Glasgow)",
    sd_our_items = sd(model_df %>% distinct(uni_lemma, familiarity) %>%
```

```

      pull(familiarity), na.rm = TRUE),
    sd_full_norms = sd(glasgow$familiarity, na.rm = TRUE))) %>%
mutate(sd_ratio = sd_our_items / sd_full_norms,
      across(where(is.numeric), ~ round(.x, 3)))
knitr::kable(rr, caption = paste(
  "Spread of the adult norms among our early nouns relative to the full",
  "norm sets: early nouns are a restricted slice of the adult lexicon."))

```

Table 3: Spread of the adult norms among our early nouns relative to the full norm sets: early nouns are a restricted slice of the adult lexicon.

norm	sd_our_items	sd_full_norms	sd_ratio
adult AoA (Kuperman)	1.652	3.044	0.543
imageability (Glasgow)	0.464	1.352	0.343
familiarity (Glasgow)	0.616	0.924	0.666

Results

(Drafted as manuscript text; update the numbers from the tables above if the data change.)

A remaining concern is reverse causality: adults may rate familiar, early, highly imageable words as shape-organized, rather than shape organization driving acquisition. We therefore refit the primary model adding adult norms that index that pathway. Adding Glasgow imageability and familiarity norms (matched for 413 unilemmas) left the shape coefficient essentially unchanged ($B = -0.34$, $p = 0.011$, against $B = -0.31$, $p = 0.0317$ in the baseline refit on the same rows). Adding adult English AoA (Kuperman et al., 2012; 588 unilemmas), itself a strong predictor of child AoA, gave $B = -0.29$, $p = 0.00852$ for shape. With all six adult covariates entered together (imageability, familiarity, adult AoA, babiness, valence, arousal), the shape coefficient was $B = -0.35$, $p = 0.0411$. On the smaller subset with babiness coverage (203 unilemmas), shape was not reliable with or without the added covariates ($B = -0.28$, $p = 0.104$ augmented; $B = -0.29$, $p = 0.0988$ baseline on the same rows), so that null reflects the reduced item set rather than the covariates. These fits do not identify the causal direction, but they are inconsistent with the specific alternative on which the shape rating merely proxies adult familiarity or the adult accessibility of early words.

Two caveats stay open and are worth stating in any response letter. First, the norms are English ratings attached to unilemmas, so they are constant across languages the way concreteness is; they control the item-level pathway, not language-specific familiarity. Second, our nouns are a restricted slice of the adult lexicon (see the range-restriction table), so the adult covariates enter with less variance than they carry in the full norm sets, and their coefficients here should not be compared with estimates from unrestricted vocabularies.